

Database model documentation

Table of contents

1. Model details	3
2. Tables	4
1.1. Table customer	4
1.2. Table technical_service	4
1.3. Table equipment	4
1.4. Table technical	5
1.5. Table appointment	6
3. References	7
2.1. Reference customer_equipment	7
2.2. Reference technical_appointment	7
2.3. Reference equipment_appointment	7
2.4. Reference technical_service_appointment	7

1. Model details

Model name:

df_SolicitudesServicios

Version:

2.4

Database engine:

MySQL

Description:

El modelo lógico representa la estructura de una base de datos diseñada para gestionar de manera eficiente el proceso de solicitud, programación y seguimiento de servicios técnicos. Este sistema está enfocado en la atención técnica de equipos de clientes, facilitando la interacción entre técnicos, clientes y servicios disponibles.

2. Tables

2.1. Table customer

Description:

Stores the personal data of customers requesting technical services. It includes contact and location information for proper service management and tracking.

2.1.1. Columns

Column name	Type	Properties	Description
client_id	int	PK	Unique customer identifier
name	varchar(100)		Client name.
lastname	varchar(100)		Client's last name.
phone	char(9)		Contact phone number
email	varchar(120)		Customer email.
document_type	char(3)		Type of identity document
ubigeo	varchar(6)		Geographic code of the customer's place of residence

2.2. Table technical_service

Description:

Define the different types of technical services the company offers. Include the name, detailed description of the service, and estimated price. This is essential for classifying and quoting services.

2.2.1. Columns

Column name	Type	Properties	Description
service_id	int	PK	Unique identifier of the technical service
service_name	varchar(100)		Service name
service_description	Text		Full details of the service offered
service_price	decimal(10,2)		Approximate cost of the service, expressed in local currency

2.3. Table equipment

Description:

Records customer-owned equipment. Each device is linked to a customer and contains information about the equipment type, brand, model, and serial number for identification.

2.3.1. Columns

Column name	Type	Properties	Description
team_id	int	PK	Unique equipment identifier
client_id	int		Unique customer identifier
equipment_type	varchar(100)		Type or category of equipment
brand	varchar(50)		Team brand
model	varchar(50)		Specific equipment model
serial_number	varchar(100)		Unique serial number of the equipment, useful for precise identification

2.4. Table technical

Description:

Contains the information of the technical staff responsible for performing the services. It includes their name, specialty, and contact information. It allows you to assign technicians to appointments based on their experience.

2.4.1. Columns

Column name	Type	Properties	Description
technical_id	int	PK	Unique technician identifier
technical_name	varchar(100)		Technician name
technical_lastname	varchar(100)		Technician's last name
specialty	varchar(100)		Specialty or technical area in which you work
technical_phone	char(9)		Technician contact number
technical_email	varchar(120)		Professional email of the technician

2.5. Table appointment

Description:

Central table that manages technical service requests. It lists the customer, equipment, technician, and requested service. It includes details such as date, type, problem description, and appointment status.

2.5.1. Columns

Column name	Type	Properties	Description
appointment_id	int	PK	Unique appointment identifier
client_id	int		Reference to the client requesting the service
technical_id	int		Unique technician identifier
team_id	int		Unique equipment identifier
service_id	int		Unique identifier of the technical service
appointment_date	timestamp		Date and time scheduled for the appointment
mode	varchar(20)		Service modality (local, home, etc.).
problem_description	longtext		Initial description of the equipment problem, provided by the customer
state	boolean		Current status of the appointment (true if active or completed, false if canceled or pending).

3. References

3.1. Reference customer_equipment

customer	0..*	equipment
client_id	<->	client_id

3.2. Reference technical_appointment

technical	0..*	appointment
technical_id	<->	technical_id

3.3. Reference equipment_appointment

equipment	0..*	appointment
team_id	<->	team_id

3.4. Reference technical_service_appointment

technical_service	0..*	appointment
service_id	<->	service_id